general-purpose computer, like a laptop, has much lesser parallelism compared to an HPC machine. The silicon chip used in general-purpose machine is designed with very strict thermal constraints (that a person can withstand) and power dissipation constraints (with which the battery must last longer).

Whereas for an HPC, although the thermal and power constraints are targeted, the biggest objective is getting maximum performance. The performance capability is measured in terms of FLOPS (floating point operations per second).

A silicon designed for high performance compute must be able to perform multiple operations in parallel to achieve higher number of compute operations in a second. The hardware architecture and compiler play a critical role in achieving such parallel processing.

The applications for HPC are much different than a general-purpose compute. An HPC application will typically have workloads as a single operation/calculation on multiple datapoints.

Hundreds of HPC silicon are arranged in parallel to create large computing machines. When such a machine is capable to perform petaflops of operations, they are called supercomputers.

Supercomputers are needed to perform highly complex calculations in a short time. The typical uses of supercomputers involve weather predictions, drug development, AI workloads, and many more.

Supercomputer's development in India

India has 23 supercomputers deployed by C-DAC at multiple locations. C-DAC develops these supercomputers through its association with various research institutes across India.

Supercomputer design and development starts with a silicon chip. The silicon chip is put on a motherboard. Multiple motherboards are connected through high-speed network cables to get parallelism needed for high-performance compute.

In 2015, National Supercomputer Mission with a budget of ₹45 billion was launched with an aim to design, manufacture, deploy supercomputers and HPC machines in India.

About 10 years ago, India did not have the capability to design and manufacture server class motherboards, so C-DAC's initial supercomputers were made using motherboards imported into India. Power and thermal designs of racks and stations were done locally so that reliability and efficiency could be achieved. The software stack was indigenised and modified for the different workloads needed for the Indian ecosystem.

In recent years, C-DAC has designed its own motherboards and is now getting them manufactured in India. The entire assembly is done in India for small production volumes. Other than the silicon chipset, which is procured from Intel, all designs are done at C-DAC for the supercomputers deployed in India.

A few months ago, Kaynes got a contract to manufacture 3000 RUDRA servers for C-DAC. India imports around 1.5 billion dollars worth of servers every year. Manufacturing these servers will bring their cost down by at least 30%-40%. This would be a big leap towards self-reliance and reducing cost, as we shall need to import only a few components rather than complete motherboards for producing a large number of servers.

More than Intel: Shift in supercomputer architecture

The silicon chipset, which is procured from Intel as of today, is the most crucial component in the high-performance compute server. Everything involved in making a supercomputer like the motherboard, the rack, cooling systems, and the software stack, depends on the architecture and features of the silicon chipset.

In the last 4-6 years, there is a trend to shift from conventional x86-based supercomputer architecture to ARM's Neoverse-based high-performance computing architecture. All major cloud operators are either designing their own high-performance compute chipset or are planning to include ARM's Neoverse-based compute engines into their portfolios.

The latest supercomputer from Japan uses Fujitsu's A64FX microprocessor, which has ARM's CPU. Since then, many more ARM-based high-performance compute processors have been released by companies like Ampere.

It is clear that there is a shift or at least additional architecture for HPC and supercomputer machines.

Bottlenecks in moving from Intel to ARM HPC platform

Although it may seem like simply swapping one processor with another, the reality is very different. Moving away from an already existing, tested and tried architecture is the reason why x86 dominated the server market for the last 20-30 years.

The compute processor comes along with its chipset, which includes interfaces, power supply, thermal management, secure boot, and many more. The compute processor cannot be replaced in isolation; the whole chipset must work in tandem with compute processor to provide 100% of functionality needed for HPC.

Apart from chipset hardware, the software created for an HPC is highly tuned for compute architecture. Over the years, the software gets matured, and any loopholes discovered are either fixed in hardware or in software. Since the architecture, chipset, and software stacks are proprietary, it is a daunting task for any new architecture to provide the same level of stability and security.